

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **Microprocessor & Computer Architecture**

Code No: **CACS155** Semester: **II**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is the use of flag? Explain different flags of 8085 with an example.
3. Draw the timing diagram of LXI H, 5070H instruction.
4. Explain how does control unit determines the type of instruction after it is read from memory.
5. What is microprogram sequencer? Differentiate between hardwired control unit and microprogrammed control unit.
6. What are the basic differences between a branch instruction, a call subroutine instruction, and program interrupt?
7. What do you mean by pipelining hazards? Explain briefly.
8. Write short notes on: a) Addressing modes of 8085 b) Vector Processing

Group C

Attempt any TWO questions [2x10=20]

9. Define symbolic microprogram. You have a list of data stored at memory location starting at 8050H. The end of the data array is indicated by data byte 00H. Write a program to add the set of readings. Store the sum at 9050 and total carry at 9051H.
10. Write a program to evaluate the arithmetic statement: $X = A - B + C * (D * E + F) / C + H * K$ a) Using a general register computer with two address instructions. b) Using an accumulator type computer with one address instructions. c) Using a stack organized computer with zero-address operation instructions.
11. Define Control Word. Explain in detail about the structure of control unit and its basic requirement.